Siddhant A. Deshmukh, PhD

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PROFESSIONAL EXPERIENCE

Data Analyst - *Max Planck Institute for Astronomy (Heidelberg, Germany)* 10/2022 - 10/2023

- Secured over €400,000 in a team of four for an entrepreneurial venture focusing on analyzing customer feedback for large conferences and trade fairs across Europe
- Developed components for online virtual networking tool using React.js
- Collected and analyzed text data using dimensionality reduction, NLP & clustering with UMAP, MDE, scipy and scikit-learn
- Led social media marketing and targeted sales campaigns increasing customer acquisition by 50%
- Created comprehensive reports for customers and interfaced with LLMs to generate key takeaways, resulting in 70% increase in customer retention

Astrophysics PhD Researcher - *Heidelberg University (Heidelberg, Germany)* 08/2019 - 12/2022

- Improved existing spectroscopic fitting routines and collaborated internationally to develop a rigorous analysis pipeline for determining the solar photospheric silicon abundance (Python)
- Performed comparisons between simulated and observed solar spectroscopic data to assess model quality
- Created codebase for coupling chemical reaction systems to hydrodynamics and radiative transfer models (Python, Julia, Fortran)
- Wrote novel analysis tools utilizing pathfinding algorithms on weighted, directed graphs to find reaction pathways for chemical networks (networkx)
- Trained convolutional neural networks to predict equilibrium chemistry, resulting in a 20x speed-up compared to directly solving the system (TensorFlow/Keras)
- Led 2 independent research projects in computational astrophysics to publication in esteemed, peer-reviewed journals

Undergraduate Researcher - *University of Exeter (Exeter, UK)* 05/2018 - 08/2018

- Analyzed spacecraft observations of the solar wind and created dashboard using numpy, scipy, matplotlib and plotly
- Optimized model parameters using Markov Chain Monte Carlo (MCMC) sampling
- Developed preliminary forecasting method for space weather with LSTM neural networks with TensorFlow/Keras, PyTorch and scikit-learn

EDUCATION

PhD in Astrophysics - *Heidelberg University (Heidelberg, Germany)* 2019 - 2023

- Computational modelling and data analysis for coupling chemical reaction systems to hydrodynamics and radiative transfer models using Fortran, Python, C, Julia
- Thesis: "Modelling Non-Equilibrium Molecular Formation and Dissociation for the Spectroscopic Analysis of Cool Stellar Atmospheres"

Integrated Master's Degree in Physics (First-Class Honours) - *University of Exeter (Exeter, UK)* 2015 - 2019

- Analysis of stellar magnetic field evolution by solving differential equations in C with data visualization in Python
- Thesis: "Refining Computational Models of Stellar Rotation Evolution Using Open Magnetic Flux"

KEY SKILLS

Programming Languages: Python (expert); Rust, JavaScript, C, Julia (intermediate); R, SQL, Java (basic)
Software: Linux, Git, GitHub, Microsoft Office, LaTeX
Languages: English (fluent), German (intermediate), Hindi (intermediate), Marathi (intermediate)

LEADERSHIP EXPERIENCE

Research Supervisor - *DAAD RISE Germany (Heidelberg, Germany)* 06/2021 - 09/2021

- Won funding for 3 months of supervising an American undergraduate exchange student in the German research environment

PhD Tutor - *Heidelberg University (Heidelberg, Germany)* 2019 - 2023

- Created material and ran tutorials for undergraduate and master's level courses in English and German including "Physics for Non-Physicists" and "Fundamentals of Simulation Methods"

Conference Presentations

- Cambridge Workshop of Cool Stars, Stellar Systems and the Sun. Toulouse, France (July 2022)
- Planetary Transits and Oscillations Stellar Science Workshop. Barcelona, Spain (Sept 2019)

Attended Schools and Workshops

- 47th Heidelberg Physics Graduate Days. Heidelberg, Germany (Oct 2021)
- International Max Planck Research Schools' Summer School on Stellar Ecosystems. Heidelberg, Germany (Sept 2021)
- KROME Summer School Workshop on Astrochemistry. Online (Feb 2021)

PUBLICATIONS

- **Siddhant A. Deshmukh**, Hans-Guenter Ludwig and Guillaume Guiglion (in prep). *Time-Dependent Molecular Chemistry in Red Giant Stellar Atmospheres with Neural Networks.*
- **Siddhant A. Deshmukh** (Oct 2023). *Modelling Non-Equilibrium Molecular Formation and Dissociation for the Spectroscopic Analysis of Cool Stellar Atmospheres*. PhD Thesis.
- **Siddhant A. Deshmukh** and Hans-Guenter Ludwig (July 2023). *Implications of Time-Dependent Chemistry in Metal-Poor Dwarf Stars.* Astronomy & Astrophysics Volume 675, A146.
- **Siddhant A. Deshmukh**, Hans-Guenter Ludwig, Arunas Kučinskas, Matthias Steffen, Paul S. Barklem, Elisabetta Caffau, Vidas Dobrovolskas, and Piercarlo Bonifacio (Dec 2022). *The Solar Photospheric Silicon Abundance According to CO5BOLD - Investigating Line Broadening, Magnetic Fields, and Model Effects.* Astronomy & Astrophysics Volume 668, A48.
- Adam J. Finley, **Siddhant A. Deshmukh**, Sean P. Matt, Mathew Owens, and ChiJu Wu. (Sept 2019). *Solar Angular Momentum Loss over the Past Several Millennia.* The Astrophysical Journal, Volume 883, Number 1.

ADDITIONAL INFORMATION

Interests: Music, martial arts, game development, creative writing
Work Eligibility: Eligible to work in the U.S. with no restrictions